Structural And Morphological Studies On Composite Electrodes Prepared By Polyol Process

P.Rajkumar, R.Subadevi, M.Sivakumar*
School of Physics, Alagappa University, Karaikudi-630 004, Tamil Nadu, India.
e-mail: susiva73@yahoo.co.in (M.Sivakumar)

***Abstract*— The $Li(Fe_xNi_yCo_z)PO_4$ (x,y,z=1/3 ; x=80%, y=10%, Z=10%) cathode materials prepared via polyol process. The as-synthesized $Li(Fe_xNi_yCo_z)PO_4$ materials characterized for structural and morphological analysis using X-ray diffraction, Fourier transform infrared spectroscopy and Scanning electron microscopy. XRD results reveals that, the prepared materials ascribed to monoclinic structure (JCPDS no : 80-1371) and lattice parameters are calculated as a = 4.822, b = 8.081 and c = 6.941. From the SEM results observed the prepared composites have been needle like structures. The Fourier transform infrared spectroscopy (FTIR) used to determine the functional group vibrations of as-prepared composite.**

Keywords—cathode, polyol process, monoclinic structure

I. INTRODUCTION

Li-ion batteries became essential for our day-to-day life, and scope of their applications is currently expanding from mobile electronic devices to electric vehicles, power tools and stationary power grid storage. Recently, olivine type $LiMPO_4$ (M=transition metal) cathode material has attractive candidate for lithium rechargeable batteries because of its low cost, environmental benignity, low toxicity, high thermal and chemical stability [1-3]. These materials typically have a high theoretical capacity (170 mAh g^{-1}) with a discharge voltage platform of 3.4–3.5 V. The strong P-O covalency stabilizes the stable nature of olivine structure, which provides good cycle life and also excellent safety during lithium ion storage [4-7].

Among these materials, particular interest towards multi-component $LiFeNiCoPO_4$ system is due to their preferred operating voltage and environmental compatibility. The advantage of multicomponent $LiFeNiCoPO_4$ system is high electronic conductivity of $LiFePO_4$, highest operating voltage of $LiNiPO_4$ and $LiCoPO_4$ [8-10]. The synthesis method consumed a significant effect on the product performance, thus played a very important role in the commercialization of these cathode materials. There are several methods used to preparing the cathode materials, they are solid state reaction, sol-gel, hydrothermal, polyol process etc., [11, 12]. Among these methods, the polyol process is an adaptable method. The polyol process is a novel method for synthesizing $LiFePO_4$ with well-defined morphologies and high crystallinity, without the need for any additional heat treatments and special atmosphere. Generally, tetra ethylene glycol (TTEG) is used as polyol solvent, it acts as a reducing agent and also a stabilizer and controls the particles growth [13].

In this present work, $Li(Fe_xNi_yCo_z)PO_4$ composite cathode material has been prepared via low temperature polyol technique with different ratio of x, y and z. The structural and morphological analysis of the prepared materials was systematically studied in detail.

II. EXPERIMENTAL

The $Li(Fe_xNi_yCo_z)PO_4$ (**B1** (x,y,z=1/3) and **B2** (x=80%, y=10%, Z=10%)) material was synthesized via low temperature polyol technique. Lithium dihydrogen phosphate (LiH_2PO_4), Nickel sulfate ($NiSO_4$), Cobalt chloride hexa hydrate ($CoCl_2.6H_2O$), Ferrous sulfate hepta hydrate ($FeSO_4.7H_2O$) were taken in stoichiometric amount, which were dissolved in tetra ethylene glycol (TTEG) and heated in a reflex condenser near to the boiling point of the polyol solvent (320 ºC) for 16 h. In order to remove the TTEG and partial organic compounds, the resulting solution was washed several times with acetone. The resulting particles were separated and dried in a vacuum oven at 150 ºC for 36 h and finally the $Li(Fe_xNi_yCo_z)PO_4$ fine particles were obtained.

III. RESULTS AND DISCUSSION

A. X-ray diffraction

Fig.1 shows the XRD patterns of the as-prepared $Li(Fe_xNi_yCo_z)PO_4$ composites B1 and B2. The structure of $LiFe_{1/3}Ni_{1/3}Co_{1/3}PO_4$ has been ascertained as monoclinic (JCPDS no : 80-1371). However, the peaks pertaining to Ni source appearing at 2θ = 20.38 and 28.28º, elucidates the incorporation of Ni into the composite. Also, peak appearing at 2θ = 35.74º corresponds to Co, which shows the incorporation of Co into the composite. Some additional peaks appearing at 2θ = 18.39, 30.90º which shows the impure phase of the composite. The cell parameters were calculated as a = 4.822, b = 8.081 and c = 6.941. The same discus-

sion is applicable for sample B2. The crystalline intensity is higher for the sample B1 than B2.

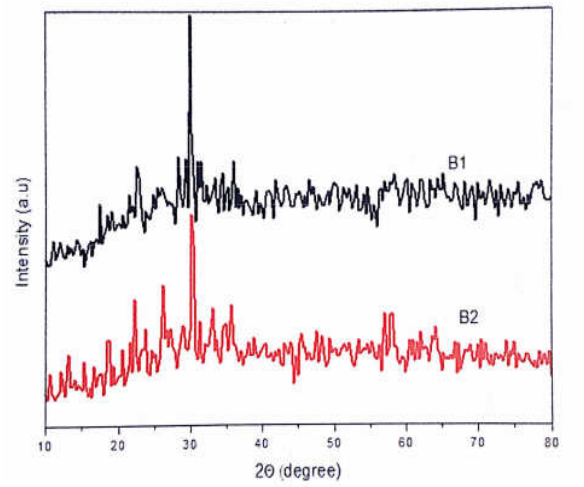

Fig. 1. XRD patterns os as-prepared samples B1 and B2

B. Fourier Transform Infrared Spectroscopy

The functional group vibrations of the prepared samples carried out using FTIR spectrophotometer. Fig. 2 shows the FTIR spectrum of as-synthesized composites B1 and B2. The broad band appears in the range of 3600 – 3000 cm^{-1} which corresponds to O-H stretching vibrations. The existence of O-H stretching is due to the entire experiment was performed at low temperature (around 325 ºC). From the fig.2 it is observed that O-H bending vibrations could be absorbed at 113 cm^{-1}. The symmetric and antisymmetric vibration O-P-O bending modes of $LiFePO_4$ have been existed at 640, 565 and 473 cm^{-1}. In the presence study, these peaks are shifted slightly in accordance with the addition of Ni and Co in the composite.

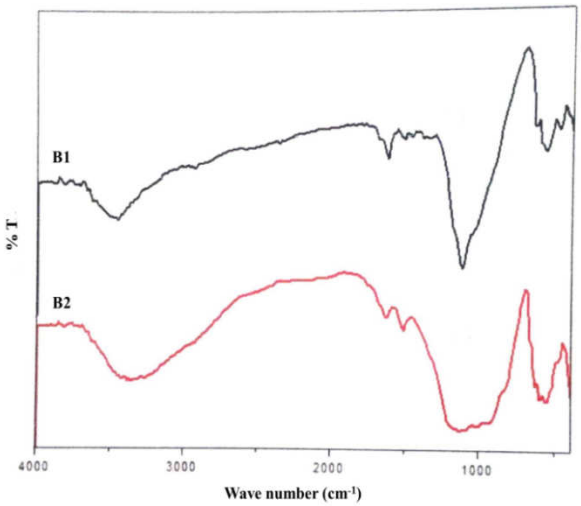

Fig. 2. FTIR spctrum of the prepared composites B1 and B2

C. Scanning Electron Microscopy

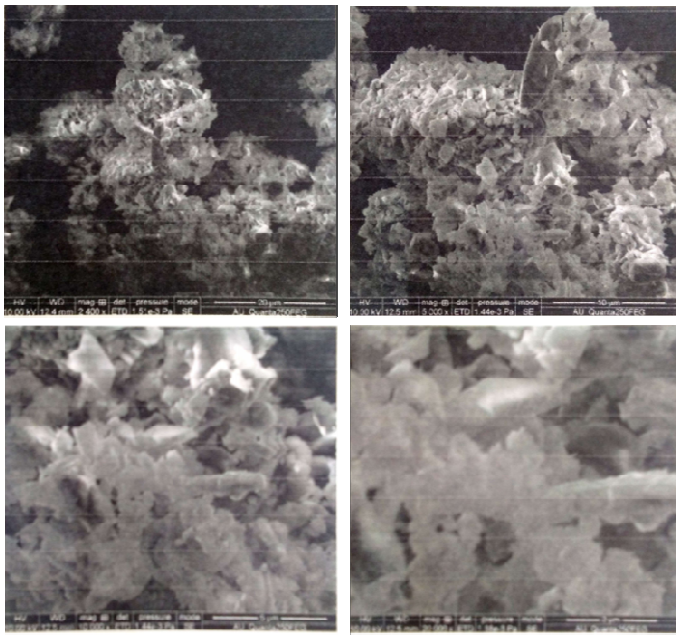

Fig. 3. SEM images of the as-synthesized composite B1

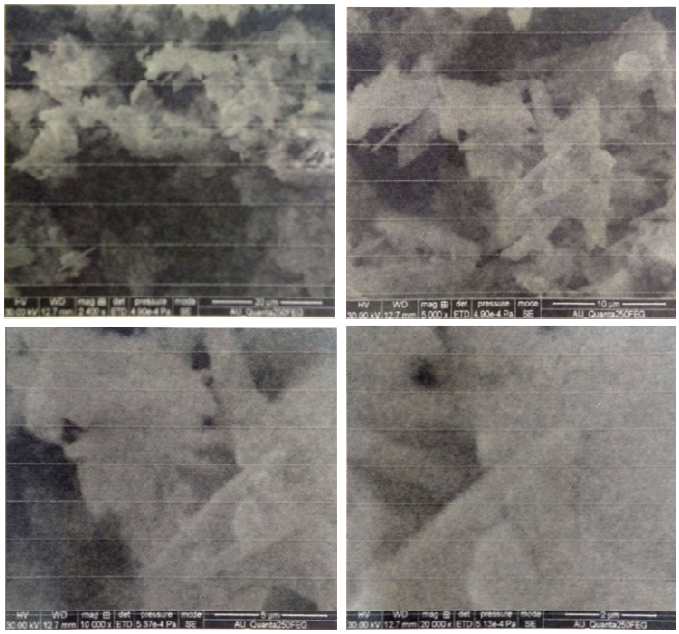

Fig. 4. SEM images of the as-synthesized composite B2

Scanning electron microscopy used to analyze the surface morphology of the prepared materials. Fig.3 shows the SEM images of the samples B1 and B2 with different magnifications. From the figure, it is clearly seen that the prepared composite is almost needle like structure with some clusters. Sample B2 consists of relatively more clusters than the other one. So, the growth of the particle is in one direction. The SEM images of the sample provide the better infor-

mation about the crystallinity in terms of the cluster formation. These results are in agreement with the X-ray diffraction analysis. From these results one can conclude that the sample B1 has the structure as required for the electrode which can be employed in the Li secondary batteries.

IV. Conclusion

An attempt has been made to synthesize the composites $Li(Fe_xNi_yCo_z)PO_4$, where x,y,z = 1/3 (B1) and x = 8/10, y = 1/10, z = 1/10 (B2) using polyol process. The prepared materials have monoclinic structure with very few impure phases. Based on the impurity and crystallinity, the equal ratio of Fe, Ni and Co based sample (B1) could be optimized among the systems studied. From SEM analysis, the needle like structure with less cluster formation of the sample is in accordance with the X-ray diffraction analysis. From these results one can conclude that the sample B1 has the structure as required for the electrode which can be employed in the lithium secondary batteries. Although, the equal ratio based sample has better crystallinity.

References

[1] M. Sivakumar, R. Muruganantham, R. Subadevi, "Investigations on rate performance of polyol technique developed $LiFePO_4/CeO_2$ composite materials for Rechargeable Lithium batteries" *RSC Advances*, 5(105):86126–86136.

[2] S. Moon, P. Muralidharan, D-K. Kim, "Carbon coating by high-energy milling and electrochemical properties of $LiMnPO_4$ obtained in polyol process" *Ceramics International*, 38S (2012) S471-S485.

[3] K-T Lee, K-S Lee, "Electrochemical properties of $LiFe_{0.9}Mn_{0.1}PO_4/Fe_2P$ cathode material by mechanical alloying" *Journal of Power Sources*, 189 (2009) 435–439.

[4] T.N.L. Doan, I. Taniguchi, "Preparation of $LiCoPO_4$/C nanocomposite cathode of lithium batteries with high rate performance" *Journal of Power Sources*, 196 (13) (2011) 5679–5684.

[5] J. Ding, Z. Su, H. Tian, "Synthesis of high rate performance $LiFe_{1-x}Mn_xPO_4$/C composites for lithium-ion batteries" *Ceramics International*, 42, (10) (2016) 12435–12440.

[6] J. Chen, "A review of nanostructured lithium ion battery materials via low temperature synthesis", *Recent Pat. Nanotechnol.* 7 (1) (2013) 2–12.

[7] J. Yao, S. Bewlay, K. Konstantionv, V.A. Drozd, R.S. Liu, X.L. Wang, H.K. Liu, G.X. Wang, "Characterisation of olivine-type $LiMn_xFe_{1-x}PO_4$ cathode materials" *Journal of Alloys and Compounds*, 425 (2006) 362–366.

[8] D. Jang, K. Palanisamy, Y. Kim, W-S Yoon, "Structural and Electrochemical Properties of Doped $LiFe_{0.48}Mn_{0.48}Mg_{0.04}PO_4$ as Cathode Material for Lithium ion Batteries" *Journal of Electrochemical Science and Technology*, 4 (3) (2013) 102-107.

[9] R. Qinga, M-C Yang, Y-S Mengb, W. Sigmund, "Synthesis of $LiNi_xFe_{1-x}PO_4$ solid solution as cathode materials for lithium ion batteries" *Electrochimica Acta*, 108 (2013) 827–832.

[10] X. Huang, J. Ma, P. Wu, Y. Hu, J. Dai, Z. Zhu, H. Chen, H. Wang, "Hydrothermal synthesis of $LiCoPO_4$ cathode materials for rechargeable lithium ion batteries" *Materials Letters*, 59 (2005) 578 – 582.

[11] R. Muruganantham, M. Sivakumar, R. Subadevi, "Synthesis and electrochemical characterization of olivine-type lithium iron phosphate cathode materials via different techniques" *Ionics*, 22(9) (2016) 1557–1565.

[12] Kim D-H, Kim JK, "Synthesis of $LiFePO_4$ nanoparticles in polyol medium and their electrochemical properties", *Electrochem Solid-State Lett* 9:A439–A442.

[13] R. Muruganantham, M. Sivakumar, R. Subadevi, "Enhanced rate performance of multiwalled carbon nanotube encrusted olivine type composite cathode material using polyol technique" *Journal of Power Sources*, 300 (2015) 496-506.